

Assessing Local Knowledge of Game Abundance and Persistence of Hunting Livelihoods in the Bolivian Amazon Using Consensus Analysis

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Abstract We used cultural consensus models to test whether hunters shared perceptions of wildlife abundance and the relative importance of hunting and fishing in a Guarayo indigenous community in the Bolivian Amazon. Results show that highly prized animals that were considered rare are either those with lower reproductive rates and more sensitivity to land use changes and harvest (white-lipped peccary and spider monkey) or those with marked seasonal distribution patterns (barred sorubim and tiger-fish). Rapidly reproducing and resilient species (agouti and armored catfish) were perceived as abundant. More tapirs and red brocket deer were present than predicted by scientific models possibly because hunters were harvesting these species in new forest management areas. Residents identified hunting and fishing among their most important livelihood activities, recognized bush meat and fish as basic food resources, and expected wildlife harvests to be part of their future livelihoods, although market-based livelihoods and domestic replacements for bush meat were reported.

Keywords Amazon · Bolivia · Guarayo Indians · Hunting knowledge · Consensus analysis

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Introduction

Sustaining faunal populations is fundamental to the well-being of many rural people in South America, especially for those whose livelihoods depend on forest resources; bush meat and fish provide families with high quality protein and fat (GDASC 2009; Ojasti 1993; Pearce 2005; Siren and Machoa 2008). In Bolivia, as in most Latin American countries, efforts to maintain forest quality are challenged by agri-business, mining, timber harvest, and commercial-fruit collecting interests (Prefectura Santa Cruz 2007; Rumiz *et al.* 2004). These operations reduce the population of wildlife species and lower success rates of subsistence hunting. Local people are increasingly drawn into the market economy fostered by commercial operations, which can lower the *need* for subsistence hunting success (Jorgenson 2000), although hunting does not necessarily decrease (Lee 2000) and may even increase with wealth (Godoy *et al.* 2010), in part for social and cultural reasons (Hill and Padwe 2000) or preference (Schenck *et al.* 2006). This article assesses how local people perceive game availability and the importance of subsistence hunting to their livelihoods.

We used a cultural consensus approach to rapidly evaluate perceptions of local hunting conditions. We tested whether hunters shared beliefs about wildlife abundance and livelihoods. Involving fisherman and hunters in data collection, specifically through self-reporting on hunts (Irvine 1987; Noss *et al.* 2004; Townsend 1995; Zapata-Ríos *et al.* 2009) and fishing trips (Copa and Townsend 2004) has been used as a tool to engage residents in wildlife management efforts as well as for the collection of the biological information required for population monitoring. These efforts capture etic (scientific) measures of wildlife, but do not address emic (local) perceptions and agreement

on problems related to wildlife. Since subsistence hunters know their hunting grounds, their perceptions on game availability and population abundance usually directly reflect results of the scientific models such as Bodmer (1994) and Robinson and Redford (1991) and are likely good indicators of actual faunal populations (Pailler *et al.* 2009). Local or traditional knowledge, of course, may not coincide with scientific knowledge. We use cultural consensus analysis as a way to systematically compare local knowledge and scientific knowledge about wildlife. We discuss how well local knowledge about game availability reflects scientific knowledge (Harris 1990; Pike 1990). For data collection we first solicited perceptions of game availability and then gauged whether hunting was important to informant livelihoods and if informants expected to follow similar livelihood strategies in the future.

The Study Area

Approximately half of Bolivia's landscape (53.1 million ha) is still covered by tropical forests, yet it is among the poorest countries of Latin America (FAO 2000). The 1996 agrarian reform law (Ley INRA no. 1715) created a new type of communal property called *Tierra Comunitaria de Origen* (TCO).¹ The law legalized the rights of indigenous communities to use forest resources for subsistence purposes within their TCO; communities with approved management plans can also commercialize forest resources. Development agencies and NGOs quickly promoted sustainable forest management in the TCOs as an alternative to forest conversion through conventional high-grade logging. Commercial timber management was intended to provide additional income and a better defense of indigenous territorial claims by demonstrating an economic use of the forests. The research described in this paper was designed to help a USAID project, the Bolivian Sustainable Forestry Project (BOLFOP I), and the Adaptive Collaborative Management Program of the Center for International Forestry Research (CIFOR) address the incorporation of hunting into forest management plans, beginning with the Guarayos' TCO, and specifically with the forest management plan of the Guarayo village Salvatierra.

Salvatierra (15°30'08"S, 63°01'33"W) is a small forest-dependent village about 360 km northeast of Santa Cruz de la Sierra on what was the northern edge of human settlements in the Guarayos' TCO in the transitional forests that give way to the Beni grasslands (Fig. 1). The landscape

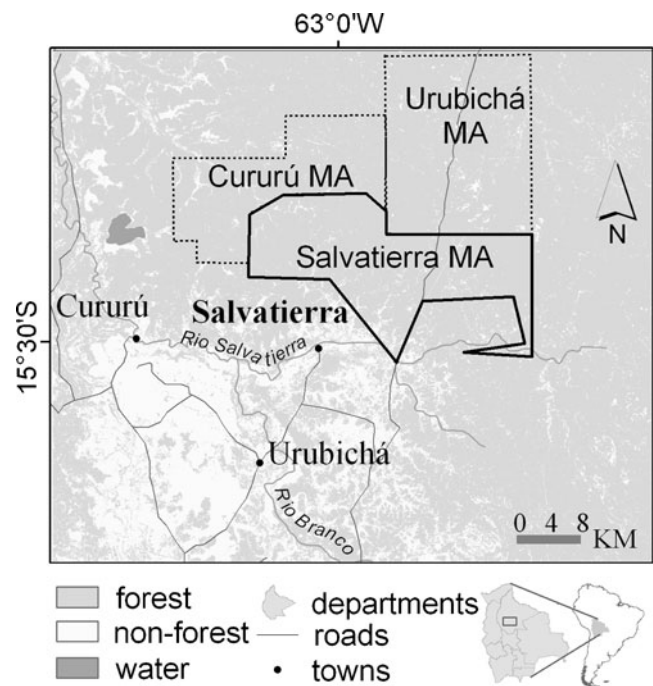


Fig. 1 Study site, Salvatierra, Bolivia, surrounding villages and the location of the indigenous forest management areas (MA). Land cover derived from 2000 and 2001 Landsat ETM+ images (path 230 & 231, row 70 & 71). Location of management areas & roads, obtained from BOLFOP in 2001. Datum: WGS 84, Projection: Geographic

is an ecotone of Amazonian and Chiquitano dry forests (Beck *et al.* 1993; Navarro and Ferreira 2008) with a mean temperature of 25°C and an average annual precipitation of 1,200 mm, concentrated in a rainy season from November to April (Toledo and Salick 2006). Flooding of the Salvatierra and Blanco rivers isolates the village of Salvatierra during high waters and makes trade difficult. Due to its remote location, Salvatierra residents have neither electricity nor potable water, and only recently received telephone service.

Salvatierra is a poor community in the poorest municipality in the Santa Cruz Department (UDAPE 2001). Settled originally in the 1930s as a multi-ethnic village within a network of Franciscan Missions, the community persisted after the missions were secularized in the early 1950s, with residents developing a livelihood system based on swidden agriculture supplemented with hunting and fishing (Stearman 1987). The villagers' diet is predominantly starch with rice, corn, manioc, and cooking bananas as staples. Beans are viewed locally as a famine food consumed only under extreme conditions. Some families also produce peanuts, but primarily as a cash crop rather than for local consumption. Hunting and fishing supply important protein resources. Starting in the 1980s, external pressure on Salvatierra's game resource base increased as the state granted logging rights to companies that practiced selective logging of stands of mahogany (*Swietenia macrophylla*) and Spanish

¹ These have been converted in the new Bolivian Constitution approved by Referendum in 2009 to TIOCs (Territorio Indígena Originario y Campesino).

cedar (*Cedrella odorata*) in the Guarayos' forests. Logging operations in lowland Bolivia customarily did not supply meat to their employees, but did supply them with bullets (Townsend *et al.* 2002). These early waves of timber extraction practically wiped out stocks of precious timber without providing significant direct benefits to the local people (Vallejos *et al.* 1996) and according to community members, decreased their hunting success. The logging roads built to extract the trees exposed the region to increased pressure from outsiders, including commercial hunters in motorized vehicles and others wishing to harvest timber illegally (Cronkleton 2005).

With the support of the BOLFOR I and CIFOR's Adaptive Collaborative Management Program, Salvatierra residents began developing a forest management plan in 2001 for a 40,886 ha area approximately 10 km north of the village. Two other indigenous community forest management areas in the Guarayos' TCO (Urubichá and Cururú) were established nearby (Fig. 1).

Methods

Sampling Design

At the time of this study (June to August 2001) 333 people (59 families) lived in Salvatierra with 51% of the population under 14 (Cronkleton 2005). We interviewed the male and female heads of 25 families, beginning with known hunters, and then used stratified snowball sampling to identify hunters in the northern, eastern, southern, and western sections of the village. The men and women from each selected household were both present for the interviews, but the men usually answered questions about fauna and hunting, while women answered questions about diet, and both answered questions about their livelihood. If the informant did not speak Spanish or needed clarification on specific points, our assistant, Hildoberto Etea, asked the questions in Guarayo, his native language.

Interviews

To assess perceptions of wildlife and fish abundance, we modified a pile sorting technique by using laminated images of 38 wildlife and 15 fish species which the informant manipulated on a felt board with Velcro. The images were obtained from well-known field guides (Eisenberg and Redford 1999; Emmons 1997; Lauzanne and Loubens 1985) and reports on local resources (Rebolledo 1995; Townsend 1995). As we went through the exercise, we discussed the name of each animal or fish to make sure that each informant understood which species was represented on each laminated image. We asked

informants to select the rare animals and then to select the abundant species. Animal images that were not selected by informant as either rare or abundant were grouped in one section of the felt board, and we asked further questions about characteristics of that section. This interactive technique eliminated confusion over multiple names for the same species and reduced communication problems stemming from language barriers (some people only spoke Guarayo) and literacy levels.

To identify the perceived relative importance of different livelihood activities we asked informants to rate the importance of each of the following: (1) hunting, (2) commercial logging, (3) firewood collection, (4) plant-based non-timber forest products (NTFPs), (5) fishing, (6) agriculture, (7) preparing food for the home, (8) selling products in the market, (9) making crafts, (10) raising poultry, (11) raising pigs, (12) raising goats, (13) raising cattle, and (14) growing fruit trees. This list of livelihood activities was developed with community members; local artists drew representations of them on 7×5 cm sheets of paper, which were then laminated onto cardstock. We discussed these cards during the interviews, making sure that informants understood which livelihood was represented by each image. We then asked informants to select and rank their five most important livelihood activities. We ranked only their top five activities because informants were not comfortable ranking activities in which they did not participate. We repeated the ranking exercise three times: (1) once for the current importance of each activity (present); (2) once for the importance of each activity for the informant's parents (past); and (3) once for the potential importance of each activity for the next generation (future).

We followed these tasks with questions about the importance of hunting for the local diet and to verify the ranking responses. First we asked informants how often they ate domestic meat, bush meat, vegetables (tomato, carrots, lettuce, etc.), fruit, and milk. We then asked informants when they had last caught bush meat; what animal they caught; how far they travelled; and what they considered to be the most delicious animal to eat. Answers were tabulated and averaged for the community.

Analysis

Cultural consensus theory is based on the assumption that, since culture is shared knowledge, people who know the answer to a question tend to agree with one another, while people who do not know the answer, and thus guess, are less likely to agree with the others. Therefore agreement among informants or consensus can be considered a proxy for knowledge if (1) responses are solicited independently

(i.e., not in a focus group); (2) only one domain of knowledge is tested at one time (a test of knowledge about modern German history and algebra would not be appropriate); and (3) there is, in fact, one set of “correct” answers to the questions (Romney *et al.* 1986; Weller 2007). Ideally, informants are asked a series of true/false or multiple choice type questions. The informant-by-informant agreement matrix is then factored. If the ratio of the first to the second eigenvalue of the factor solution is at least three-to-one (meaning that the first factor explains two thirds or more of the variance in the matrix), it indicates a cultural consensus for the responses in that domain of knowledge. A consensus analysis is preferred over simply summing up informant responses because it is statistically robust as it is based on factor analysis. For true-false and multiple-choice data, the formal model also provides an answer key—the “culturally correct” answers to the questions in the test, the answers that most informants agree is correct—adjusted for guessing. For other kinds of data, the informal model provides an answer key as the average of the values for the responses among all informants.

For the wildlife data, animals identified by the informants as rare were coded as 1, abundant animals were coded as 3, and all others were coded as 2. In seven instances, because we did not use a true pile sort, the informant incorrectly selected the animal as both rare and abundant and these cases were recoded as 2. The data on the five most important livelihoods (of the 14 listed) were coded from 1 to 5, with 5 being the most important and 1 the least important. A zero was recorded for all other activities. We ran the informal consensus model in UCINET (Borgatti *et al.* 2002) for both the livelihood and wildlife data because the data are ranked.

Results

Informants shared perceptions about the availability of wildlife and fish since the ratio of the first factor to the second factor was greater than three. They also agreed that the agouti (*Dasyprocta variegata*) was the most common animal (Table 1) and that three of the 38 species we asked about were rare: white-lipped peccary (*Tayassu pecari*), spider monkey (*Ateles chamek*), and red brocket deer (*Mazama americana*). All other animals were classified as neither abundant nor rare. Regarding local fish, informants agreed that the armored catfish (three species: *Callichthys callichthys*, *Hoplosternum littorale*, and *H. thoracatum*) were abundant and that the barred sorubim (*Pseudoplatystoma fasciatum*) and tiger-fish (*Hoplias malabaricus*) were rare (Table 1). All other fishes were considered neither abundant nor rare according to the opinions analyzed in this consensus model.

There was little variation in the informants’ ranking of the top five activities across time (Table 2). Informants agreed that agriculture, fishing, hunting, poultry production, and preparing food for the home ranked as the top five livelihood activities in present, past, and future scenarios (ratio of first to second factor was greater than 3). Hunting and fishing were important protein and fat sources (fishing ranked higher). Fish were reported as the most frequently eaten animal protein (on average four times a week), as compared to bush meat (two to three times a month). On average, informants reported that families consumed domestic meat only ten times a year making bush meat (24/year) a more important source of protein/fat than domestic meat. Milk and vegetables were reported to be consumed only a few times per year, and thus contribute little protein and fat to the diet. Raising domestic animals (pigs, goats, and cattle) was not among the most important livelihoods. Poultry production, however, was an exception, and with some people it ranked first and may signal a growing dependence on domestic products. Swidden agriculture, with a focus on manioc and corn, ranked second among the livelihoods in the past, present, and future scenarios after fishing, and before hunting, although in the present scenario, agriculture was deemed slightly more important than fishing. Collecting firewood and raising pigs also were reported as important by individuals, but did not figure among the consensus top five activities. Individuals also forecasted a growing importance in the future or a renewed importance for other market-based activities including plant-based NTFPs, commercial forestry, raising cows, making handicrafts, growing fruit trees, and selling items in the market. Though these activities did not rank in the collective top five, their presence on the list produced by the community suggests that some individuals are integrating into local and global markets.

Regarding their last hunt, 12 informants reported having most recently harvested collared peccary (*Pecari tajacu*) and 11 reported their most recent take as agouti (*Dasyprocta variegata*). Tapir (*Tapirus terrestris*) and red brocket deer (*Mazama americana*) were reported by seven informants each, paca (*Cuniculus paca*) by six informants, and two informants mentioned tortoises (*Geochelone* spp.). We did not control for group hunts so it is possible that large animals were shared, especially since we observed sharing of tapir during our study. On average, hunters reported traveling 6.35 ± 2.9 km to their most recent game capture. The hunting had occurred between one to 90 days prior to the interview, with a median of 10.5 days.

People reported consuming fish multiple times per week, bushmeat a couple of times a month, domestic meat less than once a month, and non-starch vegetables rarely (Table 3).

Table 1 The culturally correct responses of wildlife abundance (3 is abundant and 1 is rare) in Salvatierra, Bolivia, 2001 ($N=25$)

Scientific Name	Spanish	Guarayo	English	Score
WILDLIFE				
Marsupials				
<i>Didelphis</i> sp.	carachupa	mbicru guaso	opossum	2.00
Edentata				
<i>Bradypus variegatus</i>	perezoso	ai	three-toed sloth	2.00
<i>Dasybus novemcinctus</i>	tatú	tatu pira	nine-banded armadillo	1.66
<i>Euphractus sexcinctus</i>	tatú	tatu ete	six-banded armadillo	2.00
<i>Myrmecophaga tridactyla</i>	oso bandera	kaguarer	giant anteater	2.00
<i>Priodontes maximus</i>	pejichi	tatu guasu	giant armadillo	2.00
<i>Tamandua tetradactyla</i>	tamandúa	kaguarer	tamandua	1.95
Primates				
<i>Alouatta caraya</i>	manechi negro	karaya	black howler monkey	1.60
<i>Alouatta seniculus</i>	manechi colorado	karaya	red howler monkey	2.00
<i>Aotus azarae</i>	mono cuarto ojos	cuatro ojos	night monkey	2.03
<i>Ateles chamek</i>	marimono	ka'i guaso	spider monkey	1.25
<i>Cebus albifrons</i>	mono martin	ka'i ai	white-fronted capuchin	2.24
<i>Saimiri sciureus</i>	mono amarillo	ka' i poyh	squirrel monkey	2.00
Carnivores				
<i>Cerdocyon thous</i>	zorro	aguara	crab-eating fox	2.00
<i>Eira barbara</i>	melero	e'rar	tayra	2.00
<i>Puma concolor</i>	león or puma	yaguar pita	puma	2.00
<i>Leopardus pardalis</i>	ocelote	mbarakaya	ocelot	2.00
<i>Leopardus wiedii</i>	gato onza	mbarakaya	margay	1.96
<i>Puma yagouaroundi</i>	gato gris	mono michi	jaguarondi	2.00
<i>Lutra longicaudis</i>	lobito de río	kwairaka guaso	otter	2.00
<i>Mustela frenata</i>	comadreja	—	weasel	2.00
<i>Nasua nasua</i>	tejón	kuachi	coati	1.79
<i>Panthera onca</i>	jaguar or tigre	yaguar	jaguar	1.66
<i>Procyon cancrivorous</i>	zorrito	kuachi yaguar	raccoon	2.00
Perisodactyla				
<i>Tapirus terrestris</i>	anta	maenbiar rusu	tapir	2.10
Artiodactyla				
<i>Mazama americana</i>	guaso	mbaire guaso	red brocket deer	1.45
<i>Mazama gouazoubira</i>	urina	mbaire mi	grey brocket deer	1.94
<i>Tayassu pecari</i>	tropero	tayasu	white-lipped peccary	1.00
<i>Pecari tajacu</i>	taitetú	ta'etetu	collared peccary	1.85
Rodentia				
<i>Cuniculus paca</i>	jochi pintado	pal	paca	2.13
<i>Coendou prehensilis</i>	puercospin	'kuandu	porcupine	1.98
<i>Dasyprocta variegata</i>	jochi calucha	akuchi	agouti	2.88
<i>Hydrochaeris hydrochaeris</i>	capiguara	capi'guar	capybara	1.95
<i>Sciurus</i> sp.	ardilla	kuchiguai	squirrel	2.03
<i>Sylvilagus brasiliensis</i>	taití	—	rabbit	2.00
Superfamily Muroidea	ratón	saviña	rat	2.00
Order Chiroptera	murciélago	—	bat	2.00
<i>Geochelone</i> spp.	peta de monte	—	tortoise	1.91
FISHES				
Characiformes				
<i>Hoplias malabaricus</i>	bentón	tare'††	tiger fish	1.24

Table 1 (continued)

Scientific Name	Spanish	Guarayo	English	Score
<i>Leporinus trimaculatus</i>	boga	–		1.88
<i>Serrasalmus nattereri</i>	piraña	piräñ	piranha	2.37
<i>Triportheus angulatus</i>	sardina	–	–	1.92
<i>Hoplerthrinus unitaeniatus</i>	yayú	–	aimara	1.95
<i>Rhaphiodon vulpinus</i>	machete	tamokëchĩ	biara	1.85
<i>Hydrolycus scomberoides</i>	cachorro	gue	payara	1.92
<i>Prochilodus nigricans</i>	sabalo	kirimbata	black prochilodus	1.69
<i>Potamotrygon</i> sp.	raiya	yabebru	stingray	1.86
Siluriformes				
<i>Callichthys callichthys</i>	cimbao	tambata	armored catfish	2.79
<i>Pimelodus clarias</i>	bagre	yundi'a	upside-down catfish	1.86
<i>Ageneiosus brevifilis</i>	torito	duya	driftwood catfish	1.69
<i>Pseudoplatystoma fasciatum</i>	surubí	–	barred sorubim	1.03
Perciformes				
<i>Cichla orinocensis</i>	tucunare	–	peacock bass	1.82
Synbranchiformes				
<i>Synbranchus marmoratus</i>	anguila	busu	marbled swamp eel	1.60

Discussion

Wildlife Abundance Patterns

The consensus analysis easily identified species the Salvatierran hunters agreed were rare and abundant and these results, in general, coincide with those predicted by Neotropical wildlife harvest models and other ecological research; the species abundance patterns, in general, reflect those of communities where wildlife are heavily hunted and affected by land use change. These scientific models predict that vulnerability to over-hunting is related to a species' intrinsic rate of increase (how quickly an animal is replaced in the population) and body size (Robinson and Redford 1991). Typically, large animals and those with low intrinsic rates of increase, (few offspring, longer gestation, and inter-birth periods) are among the first to disappear around hunting villages (Bodmer *et al.* 1997; Robinson and Redford 1991). Similarly, large and carnivorous fishes can be the most sought-after and disappear from highly fished food webs (De Jesus and Kohler 2004; Pauly *et al.* 1998). However, other researchers suggest that the local abundance of game species is dependent first on habitat availability and access to food items, and second on vulnerability to hunting (Ojasti 2000; Shaw 1985). Disease is also theoretically a factor that explains animal abundance (Fragoso 2004), as well as source-sink dynamics² (Naranjo

and Bodmer 2007), migration patterns (Coronel *et al.* 2004), and inter specific and intra specific competition.

More than two-thirds of the informants agreed that agouti and collared peccary were the most abundant game species in the Salvatierra hunting area, reflecting the responses that these two species comprised the majority of the recently harvested game. These species are quite resistant to hunting (Painter *et al.* 1995; Peres 1996), have generally high to moderate intrinsic rates of increase (r_{max} —1.10 for agouti and 1.25 for collared peccary; Robinson and Redford 1991)—and often use garden plots as part of their foraging strategy (Jorgenson 1993). These species are frequently hunted by most forest dwelling lowland Bolivian indigenous peoples, such as the Sirionó, approximately 50 km to the west of Salvatierra (Townsend 1995; Townsend and Rumiz 2003). The Sirionó census of these two species found high densities of collared peccaries (13/km²) and agouties (26/km²) even after more than 50 years of subsistence hunting in the same area (Ino *et al.* 2001).

The agreement between informants that white-lipped peccaries are rare in the Salvatierra area may reflect hunting pressure and their low intrinsic natural rate of increase (r_{max} =0.84; Robinson and Redford 1991) or their ecology. The wide ranging migration behavior of the white-lipped peccary herds makes their appearance in any area a cause for celebration. Informants report that the white-lipped peccaries used to move through in herds of up to 200+ individuals, but these have become rare. Larger groups may actually be various smaller groups that have merged together to migrate (Kiltie and Terborgh 1983) or to use

² Source sink dynamics is a population redistribution model where wildlife production from un-hunted areas (sources) repopulates hunted areas (sinks).

Table 2 Culturally correct answers for rank order of livelihood activities (5 is most important, 1 is least important, and 0 is not important) from respondents ($N=25$) in Salvatierra, Bolivia in 2001

	Past	Present	Future
Raising poultry	2.51	3.50	2.90
Agriculture	3.49	3.38	3.44
Fishing	3.56	3.24	3.57
Hunting	2.91	2.33	2.54
Preparing food for the home	1.14	1.40	1.13
Firewood collection	0.49	0.85	0.34
Raising pigs	0.44	0.29	0.26
Making handicrafts	0	0.02	0.16
Growing fruit trees	0.25	0	0.06
Plant-based NTFP's	0.20	0	0.05
Raising cattle	0	0	0.33
Commercial forestry	0	0	0.01
Selling products in the market	0	0	0.10
Raising goats	0	0	0

abundant resources (Borman, R., [pers. comm.](#); Fragoso 1994). Scientists have postulated various causes for the decline in white-lipped peccary populations; herds have been reported as reduced by commercial hunting (Painter *et al.* 1995; Peres 1996; Reyna-Hurtado and Tanner 2007), removal of important food resources (Keuroghlian and Eaton 2008), and diseases from domestic animals (Fragoso 2004). The rarity of white-lipped peccary in the eyes of the informants may be due to the Guarayos' forest management plans or local use patterns as they relate to hunting pressure, disease, and reduction of important food resources, such as commercial harvesting of wild palm (*Euterpes* sp.) hearts to the east of Salvatierra. The actual reason is probably a combination of all these explanations.

The spider monkey (*Ateles chamek*) was perceived as rare by the informants. This species is culturally important in Salvatierra as they are consumed in celebratory feasts. It not only has a smaller body size compared to hunted ungulates, but also as a primate, it has a very low intrinsic rate of increase ($r_{max}=0.07$; Robinson and Redford 1991) making it more vulnerable to hunting. Population density estimates have been lower in hunted sites compared to un-hunted sites (Endo *et al.* 2010). The species is relatively rare

in most forests, with some exceptions in protected areas (Wallace *et al.* 1998). Spider monkeys use their habitat based on keystone food resources and will select lower quality forest types over presumably higher quality, tall forests according to the presence of favored food resources (Wallace 2004, 2006). Populations of *Ateles geoffroyi* have been shown to be indirectly affected by logging because their key resource plants do not recruit well (Gutiérrez-Granados and Dirzo 2010). Spider monkeys are shy of human activity, and thus will generally leave a logging area if they are not killed, especially if keystone resources are removed. Since this primate is probably the species most influenced by hunting and logging, it is logical that the informants agreed that spider monkeys are rare.

One indication that the hunting zone of Salvatierrans is still productive is that the seven informants reported that the tapir was the last species they hunted and that tapir were not rare. Since we did not test for group hunting, it is possible that all seven informants reported the same animal. However, the fact that a tapir was hunted and was not categorized as rare by the informants suggests that catching tapirs is not unusual. The tapir has a very low intrinsic rate of increase ($r_{max}=0.20$; Robinson and Redford 1991), has a very long life, and is late arriving to maturity making for slow population growth. At the same time, its large size (up to 200 kg) makes it a valuable catch. Tapir populations in the Bolivian dry forests are vulnerable to overhunting, yet they can have surprisingly high densities in some dry forests if not hunted (Noss *et al.* 2003). They can also reportedly withstand land use changes associated with logging if they are not hunted (Zimmerman *et al.* 2001). Another study has reported higher presence of tapirs in hunted sites (Reyna-Hurtado and Tanner 2007), but the high abundance levels are possibly due to source-sink dynamics.

The perceived red brocket deer rarity can also be related to both hunting and behavioral ecology such as source-sink dynamics. The low intrinsic rate of increase ($r_{max}=0.40$; Robinson and Redford 1991) could make red brocket deer vulnerable to hunting; however, Reyna-Hurtado and Tanner (2007) have mentioned that red-brocket deer can withstand hunting pressure. This is possibly due to multiple gestations per year, as was observed in nearby Sirionó territory about 100 km to the west (Townsend 1995). Red brocket deer abundance is also dependent upon the availability of other preferred game species (Hurtado-Gonzales and Bodmer

Table 3 Reported dietary information from informants ($N=25$) in Salvatierra, Bolivia in 2001

Consumption Reported	Minimum	Maximum	Mean	SD
Bush meat per month	0	4	1.6	1.2
Domestic animals per year	2	36	10.3	8.2
Fish per week	0	7	4.1	2.1
Vegetables (non starch) per year	0	3	1.7	1.0

2003) as well as source-sink dynamics (Naranjo and Bodmer 2007). The Salvatierran forest is a transitional area between Amazonian species from the north and Chiquitano dry forest species from the south, where grey brocket deer are reportedly more abundant than red brocket deer (Rivero *et al.* 2004). Red brocket deer are elusive, inhabiting undisturbed forests, and in northern Santa Cruz and the Beni are rarely found in ecotone and fragmented forests like those closest to Salvatierra. Even though informants perceive red brocket deer as rare, they were the fourth most common species harvested, suggesting that forest management activities may have made this species more accessible to hunters than expected from scientific knowledge.

But mere presence of tapir or red brocket deer in Salvatierran game harvest records may not accurately reflect the status of game animals in a hunting zone with a radius of 5–6 km of the community (Townsend 1995). The fact that the average reported distance traveled to where the game animals were killed was more than 6 km suggests that the majority of the game reported during this study may have been taken during work for the timber management plan, which is 8–10 km from town (Fig. 1). Informants reported that 42% of the most recently hunted animals were captured 8 km or further away. The Salvatierra workers carry firearms and will take advantage of the morning and evening foraging activities of the animals as they walk to and from work.

Informants agreed that barred sorubim and tiger-fish were rare; these species are both carnivores and important game fish in some areas; however, their perceived rarity is probably because the interviews took place during the dry season. Although both species can survive in low-oxygen waters (Coronel *et al.* 2004; Rantin *et al.* 1992), during the dry season they migrate into larger rivers (Coronel *et al.* 2004) accounting for their perceived rareness at Salvatierra. The Salvatierra environment is similar to the Sirionó area, which is an isolated landscape spotted with drying water holes during the dry season. Townsend (1995) found the tiger-fish contributed less than 1% and the barred sorubim accounted for less than 0.1% of the Sirionó catch during the dry season when the rivers and other bodies of water are diminished.

On the other hand, informants perceived armored catfish as abundant during the dry season at Salvatierra. The armored catfish has a “lung like” intestinal organ that allows it to use surface air when it “walks” between pools with its strong pectoral spines and when it estivates in the mud when the water finally dries up (Rebolledo 1995; Wilhelm and Weber 1983). During the dry season, the walking catfish can be abundant in the catch in both Salvatierra and Sirionó (Townsend 1995).

Nutritional Needs and Market Integration

Overall, people did not perceive major changes in their livelihoods when compared to the past. Nor did they foresee many changes in the future in the top five activities, although some of the activities excluded in the past and present scenarios were mentioned in the top five in the future scenarios by at least some informants. The relative stability of the rankings of the top five activities suggests that community members were not envisioning major changes and as of yet do not feel threatened by decreasing wildlife populations to a point that they do not consider the activity worthwhile in the future. This is supported by the fact that only three species of game animals were ranked as rare. Although some individuals mentioned market livelihoods, overall the community remains dependent on forest resources.

Almost every informant expressed concern about the scarcity of game and fish and viewed hunting and fishing as important livelihood activities. Informants blamed what they perceived as declining wildlife availability on factors relating to market integration (previous timber operations in the 1980s and their own harvest of animals and fish) as well as the increase in human population. Informants also reported that fishing was in many cases more dependable than hunting because you could almost always catch a fish, even if it was small, whereas you could not always catch bushmeat. Since the village is located on the banks of the Salvatierra River, most people, including women, children and the elderly, can usually catch fish.

If fish and wildlife populations decline significantly the villagers will need to grow more domestic animal protein or to trade or sell agricultural resources to purchase protein to avoid becoming vulnerable to malnutrition. The informants' first-place ranking of poultry production may signal 1) that poultry is beginning to replace or supplement fish and bush meat as a protein resource, 2) that poultry can be sold for cash in emergencies, or 3) that people are hopeful that poultry can supplement their protein/fat requirements. Another possibility is that informants may have been influenced by an NGO project that had recently distributed chickens to Salvatierra families. This project began shortly after we began our study, so poultry may have been less salient in earlier periods. Agricultural activities ranked high as a dependable resource. We observed that villagers visit their *chaco*, or agricultural plot, daily and corn and manioc are consumed daily. Although informants did not rank firewood collection in the top five livelihoods, most people viewed this activity as important, and it ranked sixth overall. With no other way to cook food, firewood supply crucially links forests to nutritional needs. No informants mentioned raising goats as a livelihood activity in any of the three scenarios.

Generally, women are responsible for food preparation in Guarayo households, and for caring for small livestock like poultry. Fishing is a gender neutral activity reported by most people including active participation by children and women, while hunting is a male domain. Agricultural production consists of a series of gender-specific and gender-neutral tasks. The activities emphasized by male and female informants reflected complementary gender roles in household maintenance. It is important that both men and women should be actively engaged in any future management strategies; women and children in particular could play leadership roles in fisheries management. As the importance of hunting declines and market integration continues, women may take on the primary responsibility for providing the household protein intake via fishing, domestic animal husbandry, or cash-generating market-based activities such as hammock sales.

Market-integration activities—logging, handicrafts, growing fruit trees, NTFPs, and selling products in the market—ranked in the top five for some informants. However, not all informants focused on the same market-based activity; the village is isolated from markets, and people are still dependent on forest resources. Even though at the time of this fieldwork the community did not yet have an approved forest management plan and no timber had been sold, with more than half the families investing labor to prepare the plan, it is surprising that commercial forestry was not ranked in the top five, even in the future scenario. Some women weave hammocks and others want to revitalize hammock making, but the sale of the hammocks requires stronger market connections and access to materials and equipment. Fruit, such as oranges, was reported to be consumed only in season and only a few people had fruit trees; however, oranges are sold in the market when the river is low and they can be easily transported by road to the municipal capital Urubichá. NTFPs were not considered important at present and few informants ranked them as important in the past or future. No informant mentioned selling items in the market as an important livelihood in the present, but five ranked this activity within the top five in the future.

Bushmeat Conservation and Management and Market Integration

The results of the consensus analysis were useful in stimulating a lively discussion in Salvatierra about conservation and management possibilities for wildlife. During these exchanges we learned that people were not yet willing to adopt an outright moratorium on hunting or even harvest quotas. Community members suggested that in the future if they perceive the necessity to do so, they would be willing to limit hunting for certain species, but only if everyone

agreed to comply. Hunters, in particular, were concerned that restrictions on hunting would be difficult to enforce because people must voluntarily change their behavior and respect social controls. People were also concerned that neighbors in Urubichá and Cururú villages who share game resources would also need to comply.

The informants agreed (1) on wildlife abundance patterns, (2) that hunting and fishing were both important to the communities' livelihood, and (3) that they envision that these activities will remain important in the future. In this study, the emic models of wildlife abundance correspond to the etic models for white-lipped peccary, agouti, barred sorubim, tiger-fish, and armored catfish. The etic models explain that abundance patterns reflect effects from hunting and land use change, habitat preferences, or migration patterns. The etic models did not match for the tapir and red brocket deer; these animals were perceived more abundant than predicted by scientific models, suggesting that hunters are likely harvesting these species in the forest management areas.

Systematically comparing the emic and etic models can help with wildlife management. When the models do not coincide, wildlife managers could use the results from the consensus analysis to tailor intervention programs for particular species. Based on the data from this study, people should be receptive toward a management program for white-lipped peccaries and/or spider monkeys because scientists and locals agree that these species are rare, and hunting likely plays a role in their rarity. People may not be as receptive to a management program for tapir because even though scientific models predict that tapir are rare, hunters did not perceive this rarity. Red brocket deer might also be problematic for management because even though people perceive these animals as rare, and the models predict that these species are vulnerable to hunting, ecological knowledge about the red brocket deer in this forest type would suggest that they are naturally rare rather than affected by hunting. Hunters, however, are reportedly bringing in red-brocket deer. Red brocket deer and tapir may be a good starting point to discuss source-sink dynamics and hunting in the forest management area.

The results of the comparison of emic to etic knowledge suggests that barred sorubim and tiger-fish could be introduced in a management plan that incorporates seasonal distribution patterns, although it is first necessary to assess if these species are actually abundant at other times of the year as we suspect. Finally, abundant species such as armored catfish and agouti could also be part of a comprehensive wildlife management plan as harvestable species because they resist harvest pressure.

Wildlife production, and therefore, the abundance and consumption of bush meat, is related to habitat quality, the area and extent of the hunting zone, land cover/use of areas

surrounding the hunting zone, and on hunting pressure. It also depends on human population size, wildlife characteristics (reproductive rates, health, predator and prey abundance), and the presence of alternative food resources for people, such as fish, domestic animals, and vegetables. Finally, it may also depend on the extent to which people are engaged in wage labor and/or are connected to markets.

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